

a market research report published by Market-sandMarkets, 3D printing market is expected to reach US\$ 32.78 billion by 2023.

The latest market study released by Technavio predicts that the global 3D printer market is projected to reach US\$ 11.92 billion by 2021, at a CAGR of more than 35 per cent from 2017 to 2021 (Fig. 1).

McKinsey Global Institute indicates that by 2025, 3D printing could have an economic impact of US\$ 200 to US\$ 500 billion annually. This huge market opportunity is attributed to factors like ease of development of customised products, ability to reduce overall manufacturing costs and government investments in 3D printing projects for the development and deployment of the technology.

North America holds the largest share of the 3D printing market, followed by Europe and Asia Pacific. Global market growth is driven by strong demand from A&D, healthcare, education and consumer product industries.

In India, the market for 3D printers is still at a nascent stage. However, it offers huge growth opportunities in the coming years. According to 6Wresearch, a market intelligence and advisory company, the 3D printer market

in India is projected to touch US\$ 79 million by 2021. The thrust on domestic production, demand for lean manufacturing and increasing penetration across various applications, coupled with Make in India campaign will spur the 3D printer market in India.

With initiatives like Make in India, the domestic manufacturing sector is being encouraged. This will automatically play a pivotal role in the growth of the local 3D printing industry. Considering benefits associated with 3D printing, various industries in India have already started using or are ready to use 3D printing technology. Therefore the 3D printing service industry is growing at a good pace, but the 3D printer manufacturing sector will need a strong boost.

At present, India primarily imports 3D printers from countries such as China, the US and Germany.

ago, there were almost no companies in this industry—this has increased to five to seven companies. People are getting easy access to 3D printing, and the market's expectations are high.

The rapidly growing user base of professional computer-aided design/computer-aided manufacturing (CAD/CAM) help make engineering designs in the form of digital models that can be directly used as inputs for 3D printing. Adoption of digital CAD processes in traditional manufacturing sectors has created opportunities to introduce more product lines, enabling rapid prototyping and fast manufacturing processes even for products and components with small production volumes.

3D printers offer huge cost and time reductions in certain industrial applications, while providing better accuracy compared to conventional

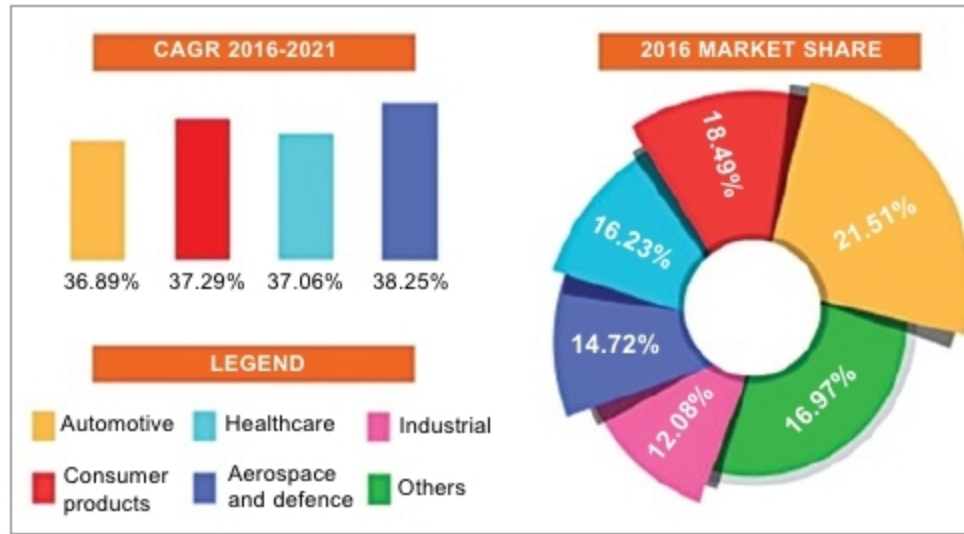


Fig. 1: Global 3D printers market growth rate by end user (Source: Technavio)

Usage of 3D printing in India

Electronics. 3D printing is used by product designers in the consumer electronics space to create better designs. Apart from this use, the electronics industry is experimenting to 3D-print electronic circuit boards using multiple extruders and conductive inks. With the advent of the Internet of Things (IoT), Raspberry Pi and Arduino, customised casings for circuit boards and other equipment are in high demand. 3D printing can create casings for these kinds of circuit boards.

Automotive. This sector is a major adopter of 3D printing globally. 3D printing is used in the automotive sector mainly for rapid in-house prototyping, producing spares for outdated vehicles and material experimentation.

However, in India, adoption is not quite aggressive, but it is moving in the right direction. Companies like Nissan and Mahindra have started using 3D printing in their production and prototyping processes.

Medical. 3D-printed prosthetics, surgical guides, dental implants and more help make treatments less complicated and more affordable. The Indian medical sector has started using 3D printing quite interestingly. For example, recently, at Gurugram's Medanta hospital, doctors implanted a 3D-printed vertebra in a woman—a third-of-its-kind surgery in the world.

Moreover, in dentistry, 3D-printed dental aligners have been adopted in

India as a popular alternative for metal teeth braces.

Architecture. Architects use 3D printing for creating miniature models of ambitious and artistic yet complex structures. The quality and intricacy produced using 3D printing is far better than any other conventional method, including laser cutting.

A&D. The A&D industry is an early adopter of 3D printing technology worldwide. Current applications include concept modelling and prototyping, and printing low-volume complex aerospace parts and replacements parts.

Emerging applications include embedding additively-manufactured electronics directly on parts, and printing aircraft wings, complex engine parts and repair parts on the battlefield.

In India, the A&D industry has started using 3D printing technology mainly for demonstrating and experimenting new designs and building prototypes. In India, GKN Aerospace is one of the major adopters of the technology.

Education. In the education sector, even though usage of 3D printing is at a formative stage, many academic institutions are starting to set up 3D printing labs. Researchers and students are using 3D printing for creating custom parts and structures for their projects and experiments.